

MN&J Labs - Command Reference

Command Center, Firebase, Profly, mail and ports. Generated 2026-08-04.

1. Ports at a glance

Port	What	Started by
5178	Command Center (generic web app)	IntelliJ run config, Server.bat, or server-ctl.ps1 start
4000	Firebase Emulator UI - browse Firestore + Auth, read function logs	firebase emulators:start
5000	Firebase Hosting - the Profly console page	firebase emulators:start
5001	Cloud Functions (the REST API lives here)	firebase emulators:start
8181	Firestore emulator (Java process). 8181 avoids the usual 8080 clash	firebase emulators:start
9099	Firebase Auth emulator	firebase emulators:start
9199	Firebase Storage emulator	firebase emulators:start
8098	Profly Kotlin API - LOCAL only (AWS still uses 8090)	java -jar / IntelliJ

Only these ports can be killed by the tooling. Anything else is refused by design.

2. Command Center (generic)

Location: C:\claude\generic - the web app on port 5178.

Control script - all actions

```
cd C:\claude\generic

.\server-ctl.ps1 status # running? which PID
.\server-ctl.ps1 start # launch if not already up
.\server-ctl.ps1 stop # graceful stop
.\server-ctl.ps1 restart # stop then start
.\server-ctl.ps1 open # start if needed + open the browser
.\server-ctl.ps1 ports # every project port and what holds it
.\server-ctl.ps1 kill # FORCE-free port 5178
.\server-ctl.ps1 killall # free 5178 AND all Firebase emulator ports
```

kill vs stop: "stop" asks nicely and can miss an orphan; "kill" frees the port whatever is holding it.

Start with mail credentials (one command - the vars must reach the same process)

```
cd C:\claude\generic; . .\mail-env.ps1; npm start
```

Since the server also parses mail-env.ps1 itself at startup, a bare `npm start` or `node server.js` now works too.

Other ways to run it

How	What to do	Notes
IntelliJ	Run config "MyTeam" (Node.js) or "Generic server" (Shell Script)	Leave Environment variables EMPTY - server.js reads mail-env.ps1
Server.bat	Double-click; menu 1-8	6 = show ports, 7 = kill 5178, 8 = kill all
In-app	Server menu in the header	Status, Restart, Stop, per-port "free", "kill all"

3. Firebase - everyday commands

Install / login (one time)

```
npm install -g firebase-tools
firebase --version
firebase login # only needed to DEPLOY, not for emulators
```

The dev loop (Proofly example)

```
cd C:\myPrograms\mnj\labs\firebase\proofly-fb

npm --prefix functions run build # 1. compile the TypeScript functions
firebase emulators:start # 2. the WHOLE backend, offline

# --- second terminal ---
npm run seed # 3. seed Firestore + Auth (idempotent)
node seed/smoke.js # 4. 15 contract + tenant-isolation tests
```

Emulator data lives in memory. Every restart wipes Firestore and Auth - re-run `npm run seed`. To keep it: `firebase emulators:start --import=./data --export-on-exit=./data`

Useful variations

```
firebase emulators:start --only functions,firestore,auth
firebase emulators:start --project demo-foo # "demo-" prefix skips all cloud calls
npm --prefix functions run watch # recompile TS on save
firebase deploy # everything (needs Blaze plan)
firebase deploy --only functions
firebase deploy --only firestore:rules,firestore:indexes
firebase functions:log
firebase projects:list
```

Where things are while the emulators run

URL	What
http://localhost:5000	Proofly console (Hosting)
http://localhost:4000	Emulator UI - data browser, auth users, logs
http://localhost:4000/firestore	Browse/edit Firestore directly
http://localhost:5001/proofly-mnjlabs/europe-west1/api/health	API health

URL	What
http://localhost:5001/proofly-mnjlabs/eur-ope-west1/api/templates	API (needs a Bearer token)

Function URL shape: `http://localhost:5001/<projectId>/<region>/<functionName>`

Demo sign-ins (emulator only)

Email	Password / org
s.braun@hausverwaltung.de	proofly123 - org-hausverwaltung (owner)
yusuf@markt-demir.de	proofly123 - org-markt-demir (owner)

Two different orgs on purpose: sign in as each and watch the report list change. That is firestore.rules doing the work, not app code.

4. Mail (Command Center -> Naum)

Check it without sending anything

```
curl "http://localhost:5178/mail/status" # configured?
curl "http://localhost:5178/mail/status?verify=1" # real SMTP login, sends nothing

# compose a message and inspect it without sending:
curl -X POST "http://localhost:5178/mail/send?dry=1" -H "Content-Type: application/json" ^
-d "{\"to\":\"x@y.com\",\"subject\":\"s\",\"body\":\"b\"}"
```

Credentials live only in `C:\claude\generic\mail-env.ps1` (gitignored). It holds a Google **App Password**, not the account password - Gmail SMTP rejects normal passwords.

If the send button says **"MAIL_USER / MAIL_PASS are not set"**, the server process was started before the credentials existed. Restart it - env vars are read at launch and never picked up later.

5. Proofly - the two backends

Stack	How to run	Base URL
Kotlin + Mongo	<code>java -jar target\proofly-api-0.1.0.jar</code>	<code>http://localhost:8098</code>
Firebase	<code>firebase emulators:start</code>	<code>http://localhost:5001/.../api</code>
AWS (live)	already deployed	<code>http://mnjlabs-alb-...elb.amazonaws.com</code>

Flutter app (Android Studio)

```
cd C:\myPrograms\mnjlabs\frontend\proofly-app
flutter devices # what is connected
flutter emulators # available AVDs
flutter emulators --launch Pixel_10_Pro

flutter run -d chrome # web, against AWS
flutter run -d chrome --dart-define=API_BASE=http://localhost:8098
flutter run -d <deviceId> --dart-define=API_BASE=http://10.0.2.2:8098
```

10.0.2.2, not localhost. Inside the Android emulator "localhost" means the emulated phone itself. 10.0.2.2 is your PC. This is the number-one reason a Flutter app "cannot reach the API".

IntelliJ/Android Studio run configs already exist: "Proofly - AWS (live)" and "Proofly - local Kotlin API".

6. Troubleshooting - things that actually bit us

Symptom	Cause and fix
Port already in use	A previous run is still holding it. <code>.\server-ctl.ps1</code> ports to see who, then <code>kill / killall</code> .
Mail says not configured	Server started before <code>mail-env.ps1</code> existed. Restart the server.
Page behaves like old code	Browser cached the HTML. <code>Ctrl+Shift+R</code> once (HTML is now served no-store).
Emulator data gone	Expected - it is in memory. Re-run <code>npm run seed</code> .
Functions "does not exist"	The functions codebase failed to LOAD. Read the emulator log, not the HTTP 404.
FieldValue is undefined	<code>import * as admin + esModuleInterop</code> . Use modular imports: <code>getFirestore</code> , <code>FieldValue</code> , <code>getAuth</code> .
Cannot find module @firebase/app	<code>npm</code> did not hoist an admin peer dep: <code>npm i @firebase/app</code> inside functions/.
Flutter cannot reach API	Android emulator: use 10.0.2.2, not localhost.
Client gets 401 after role change	ID token is stale. Call <code>getIdToken(true)</code> to refresh custom claims.
firebase deploy asks for billing	Cloud Functions need the Blaze plan. Expected; a card must be on file.